

BIOL90042 tutorial 1

1. Name the four panels of RStudio, and briefly describe what each one is used for
2. What would be the result of evaluating the following in R?

- a. `200 / 10`
- b. `4 ^ 2`
- c. `4^2`
- d. `0.1 * 6`
- e. `5/0`
- f. `1 + 2 + 3 + 4`
- g. `2 + 3 * 4`
- h. `(2 + 3) * 4`
- i. `10 + 6 0`

Solution.

2. Solutions to Q2:

- a. 20
- b. 16 (the ^ symbol means 'to the power of' in R)
- c. Also 16 (remember, white space around the mathematical operators is not required)
- d. 0.6
- e. Inf (infinity)
- f. 10 (you can chain as many mathematical operators together as you like)
- g. 14 (because of order of operations, R does 3 times 4 first then adds 2)
- h. 20 (this time, R does the addition first as it's in brackets)
- i. An error: 'unexpected numeric constant' because of the space between the 6 and the 0. Either the numbers should all be separated by mathematical operators (e.g. `10 + 6 + 0` or `10 + 6 * 0`) or there should be no space between the digits if `6 0` was supposed to be `60`

3. Write R code to solve the following mathematical problems

- a. $100 + 5$
- b. $0.2 \div 0.4$
- c. $2^2 + 3^2 + 4^2$
- d. $\frac{(15+2)^3}{100}$

Solution.

3. Solutions to Q3:

(Note: this is one possible solution. You may have code with/without brackets or white space that gives the same answer, which is totally fine).

- a. `100 + 5`
- b. `0.2/0.4`
- c. `2^2 + 3^2 + 4^2`
- d. `(15+2)^3 / 100`

The following question asks you about file paths. Please answer them according to the operating system of your computer. If there are people around you with a different operating system, try comparing your answers!

- 4. I have a file called 'tutorial_1.R' inside of a folder called 'BIOL90042_tutorials', which is located on my desktop.
 - a. Write a **relative** path to 'tutorial_1.R', from the desktop
 - b. Write an **absolute** path to 'tutorial_1.R'
 - c. Why did I name my file 'tutorial_1.R' instead of 'tutorial 1.R'?
 - d. If I told you I'd made a mistake, and the folder was actually called 'Biol90042_tutorials', would the paths you wrote for parts a and b still work? Why/why not?

Solution.

4. Solutions to Q4:

- a. On mac/linux: `BIOL90042_tutorials/tutorial_1.R` or on windows `BIOL90042_tutorials\tutorial_1.R`
- b. Exact paths would depend on your own machine, but as an example, on mac/linux: `/home/Users/BIOL90042_tutorials/tutorial_1.R` or on windows `C:\Users\YourUsername\Desktop\BIOL90042_tutorials\tutorial_1.R`.
- c. I used the underscore instead of a space because, when coding, it's much easier to avoid using spaces. Spaces in paths typically need to be 'escaped' (e.g. `BIOL90042_tutorials/tutorial\ 1.R` on mac/linux or `BIOL90042_tutorials^ tutorial_1.R` on windows), which can be annoying/difficult to read. So, it's recommended to avoid spaces wherever possible.

- d. No, the paths in parts a and b would no longer work, because they have been written for a folder called 'BIOL90042_tutorials' not 'Biol90042_tutorials', and paths are case-sensitive.
5. Challenge: I want to use R to calculate the volume of a sphere (recall that the formula for this is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere). My sphere has a radius of 3 cm.
- Using the approximation $\pi \approx 3.14$, write some R code to calculate the volume of this sphere.
 - Now, re-write the code using `pi` instead of 3.14. Are the answers the same? Why/why not?
 - Which line of code, the one from part a or part b do you think would be the best way to calculate the volume of a sphere in practice? Why?
 - `pi` is what is known as a *constant*, it's a special name for a value that won't ever change. The opposite of a constant is a variable. In the formula above, what is the variable?
 - In R, you can assign a value to a variable using the assignment operator `<-` (type `<` and then `-` on your keyboard). For example, to assign the value of 2 to a variable called `x`, you would write `x <- 2`. Use this information to re-write the code from part b to use a variable.
 - What do you think is the advantage of using variables in R?

Solution.

5. Solutions to Q5:

- `4/3 * 3.14 * 3^3`, or you could use brackets to separate the fraction like `(4/3) * 3.14 * 3^3`. Note that because R processes division before multiplication, there is no difference between the two.
- `4/3 * pi * 3^3` (literally just replace 3.14 with `pi`). The answers aren't quite the same: the code from part a gives 113.04 whereas using `pi` we get 113.0973. This is because `pi` in R has been computed to several hundred decimal points, so it gives a more precise answer.
- The code from part b, that uses `pi` would be best. This is because it is the most precise. Someone has already done the hard work of programming the value of π into R, so take advantage of this!
- In the formula for the volume of a sphere above, the radius, r , is the variable.
- We could write `r <- 3` and then `4/3 * pi * r^3`. Try running these lines of code to see that you get the same answer as for part b.
- Using variables in R allows us to more easily re-use our code. For example, if I had another sphere with radius 2 cm, I could easily re-assign the value of 2 to `r` (by writing `r <- 2`) and then re-run my volume calculation. It also makes

it clearer what calculation we're performing. If we look at the code from part a, there are a bunch of numbers, and it's not immediately obvious that we're calculating the volume of a sphere, whereas the code from part e looks a lot more like the original mathematical formula we were given. It's important to remember that code doesn't just exist to provide an answer, it should also be a readable description of what you are doing.